

Government Engineering College Patan

Computer Science and Engineering

B.E. Semester-7

Subject: Machine Learning (3170724)

Midsem Syllabus

Date:25/08/2026

Sr. No.	Content
1	Introduction to Machine Learning: Overview of Human Learning and Machine Learning, Types of Machine Learning, Applications of Machine Learning , Tools and Technology for Machine Learning .
2	Preparing to Model: Machine Learning activities, Types of data in Machine Learning, Structures of data, Data quality and remediation, Data Pre-Processing: Dimensionality reduction, Feature subset selection.
3	Modelling and Evaluation: Selecting a Model: Predictive/Descriptive, Training a Model for supervised learning, model representation and interpretability, Evaluating performance of a model, Improving performance of a model.
4	Supervised Learning: Classification and Regression: Supervised Learning, Classification Model, Learning steps, Classification algorithms, Regression, Regression algorithms,
5	Unsupervised Learning: Supervised vs. Unsupervised Learning, Applications, Clustering, Association rules